

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics

PRACTICAL I(Task)

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Class: TYIT	Batch: 1
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```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

Q.1 Write a Python program to:

- Find the Top 10 Genres.
- Find the Bottom 10 Genres.
- Identify the most popular Genre.
- Identify the least popular Genre.
- Display the result using a bar chart.

Code:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("NetFlix - NetFlix.csv")
```

```
# Split multiple genres
```

```
genre = df["genres"].dropna().str.split(",")
```

```
# Convert list into rows
```

```
genre = genre.explode().str.strip()
```

```
genre_count = genre.value_counts()
```

```
print("Top 10 Genres")
```

```
print(genre_count.head(10))
```

```
print("\nBottom 10 Genres")
```

```
print(genre_count.tail(10))
```

```

print("\nMost Popular Genre")

print(genre_count.idxmax(), "-", genre_count.max())

print("\nLeast Popular Genre")

print(genre_count.idxmin(), "-", genre_count.min())

# Bar Chart

genre_count.head(10).plot(kind="bar", color="skyblue")

plt.title("Top 10 Genres(Ritesh Chaubey)")

plt.xlabel("Genres")

plt.ylabel("Number of Titles")

plt.xticks(rotation=45)

plt.show()

```

Output:-

```

Top 10 Genres
genres
International Movies    2437
Dramas                 2106
Comedies               1471
International TV Shows 1199
Documentaries          786
Action & Adventure     721
TV Dramas              704
Independent Movies     673
Children & Family Movies 532
Romantic Movies        531
Name: count, dtype: int64

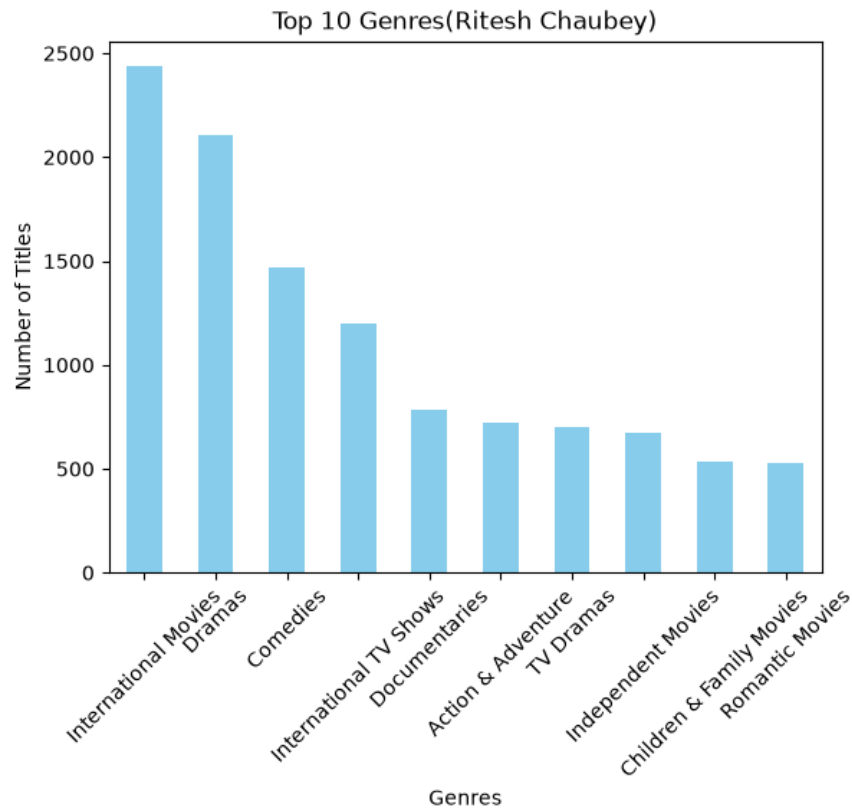
```

```

Bottom 10 Genres
genres
TV Horror                69
Teen TV Shows           60
Cult Movies             59
Faith & Spirituality    57
Anime Features          57
Movies                 56
Stand-Up Comedy & Talk Shows 52
TV Thrillers            50
Classic & Cult TV       27
...
International Movies - 2437

Least Popular Genre
TV Shows - 12

```



Q.2 Write a Python program to find:

- The oldest Movie.
- The newest Movie.
- The oldest TV Show.
- The newest TV Show.

Display all details of these titles.

Code:-

```
movies = df[df["type"] == "Movie"]
print("Oldest Movie")
print(movies.loc[movies["release_year"].idxmin()])

print("\nNewest Movie")
print(movies.loc[movies["release_year"].idxmax()])
tvshows = df[df["type"] == "TV Show"]
print("\nOldest TV Show")
```

```
print(tvshows.loc[tvshows["release_year"].idxmin()])
```

```
print("\nNewest TV Show")
```

```
print(tvshows.loc[tvshows["release_year"].idxmax()])
```

Output:-

```
Oldest Movie
show_id          s4961
type             Movie
title            Prelude to War
director         Frank Capra
cast             NaN
country          United States
date_added       31-Mar-17
release_year     1942
rating           TV-14
duration         52
genres           Classic Movies, Documentaries
description      Frank Capra's documentary chronicles the rise ...
Name: 4402, dtype: object
```

```
Newest Movie
show_id          s1286
type             Movie
title            Charming
director         Ross Venokur
cast             Wilmer Valderrama, Demi Lovato, Sia, Nia Varda...
country          Canada, United States, Cayman Islands
date_added       8-Jan-21
release_year     2021
rating           TV-Y7
duration         85
genres           Children & Family Movies, Comedies
description      On the eve of his 21st birthday, an adored pri...
Name: 319, dtype: object
```

```

Oldest TV Show
show_id          s4868
type             TV Show
title            Pioneers: First Women Filmmakers*
director         NaN
cast             NaN
country          NaN
date_added       30-Dec-18
release_year     1925
rating           TV-14
duration         1
genres           TV Shows
description       This collection restores films from women who ...
Name: 4298, dtype: object

```

```

Newest TV Show
show_id          s1223
type             TV Show
title            Carmen Sandiego
director         NaN
cast             Gina Rodriguez, Finn Wolfhard, Abby Trott, Mic...
country          United States
date_added       15-Jan-21
release_year     2021
...
duration         4
genres           Kids' TV, TV Thrillers
description       A master thief who uses her skills for good, C...
Name: 250, dtype: object

```

Q.3 Write a Python program to:

- Find the country with the highest Netflix content.
- Display the Top 10 Countries.
- Display the Bottom 10 Countries.
- Display the result using a horizontal bar chart.

Code:-

```
country = df["country"].dropna().str.split(",")
```

```
country = country.explode().str.strip()
```

```
country_count = country.value_counts()
```

```

print("Country with Highest Netflix Content")

print(country_count.idxmax(), "-", country_count.max())

print("\nTop 10 Countries")

print(country_count.head(10))

print("\nBottom 10 Countries")

print(country_count.tail(10))

# Horizontal Bar Chart

country_count.head(10).sort_values().plot(kind="barh", color="orange")

plt.title("Top 10 Countries(Ritesh Chaubey)")

plt.xlabel("Number of Titles")

plt.ylabel("Country")

plt.show()

```

Output:-

```

Country with Highest Netflix Content
United States - 3297

```

Top 10 Countries

```

country
United States    3297
India            990
United Kingdom   723
Canada           412
France           349
Japan            287
Spain            215
South Korea      212
Germany          199
Mexico           154
Name: count, dtype: int64

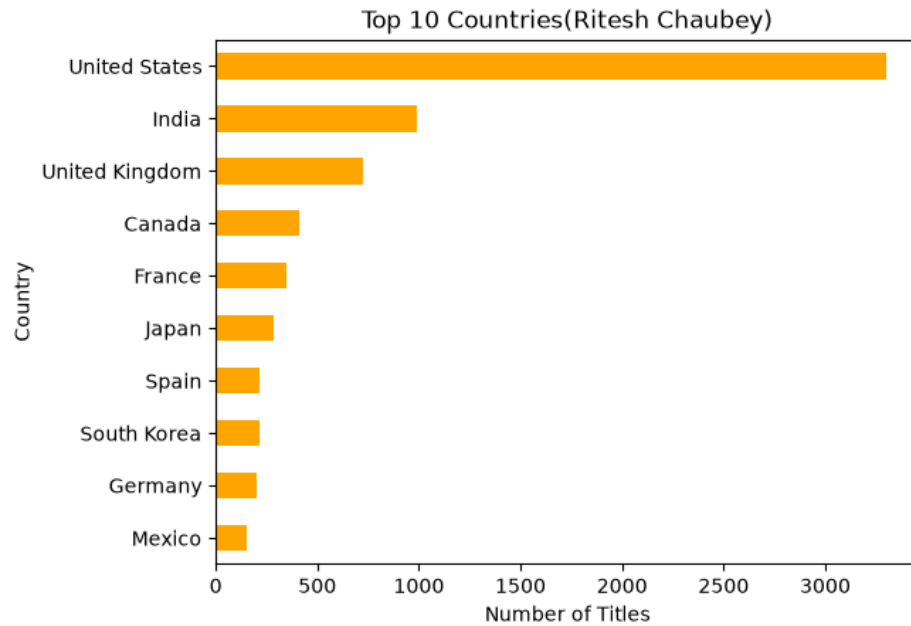
```

Bottom 10 Countries

```

country
Paraguay         1
Somalia          1
Sudan            1
Panama           1
Armenia          1
Mongolia         1
...
East Germany     1
Montenegro       1
Syria            1
Name: count, dtype: int64

```



Q.4 Write a Python program to:

- Find the director with the highest number of Movies.
- Find the director with the highest number of TV Shows.
- Display the Top 10 Directors.
- Plot a bar chart of the Top 10 Directors.

Code:-

Movies

```
movie_director = (
    df[df["type"] == "Movie"]["director"]
    .dropna()
    .str.split(",")
    .explode()
    .str.strip()
)

movie_count = movie_director.value_counts()
```

```
print("Director with Highest Movies")  
print(movie_count.idxmax(), "-", movie_count.max())
```

```
# TV Shows
```

```
tv_director = (  
    df[df["type"] == "TV Show"]["director"]  
    .dropna()  
    .str.split(",")  
    .explode()  
    .str.strip()  
)
```

```
tv_count = tv_director.value_counts()
```

```
print("\nDirector with Highest TV Shows")  
print(tv_count.idxmax(), "-", tv_count.max())
```

```
print("\nTop 10 Directors")  
print(movie_count.head(10))
```

```
# Bar Chart
```

```
movie_count.head(10).plot(kind="bar", color="green")  
plt.title("Top 10 Directors(Ritesh Chaubey)")  
plt.xlabel("Director")  
plt.ylabel("Number of Movies")  
plt.xticks(rotation=45)
```



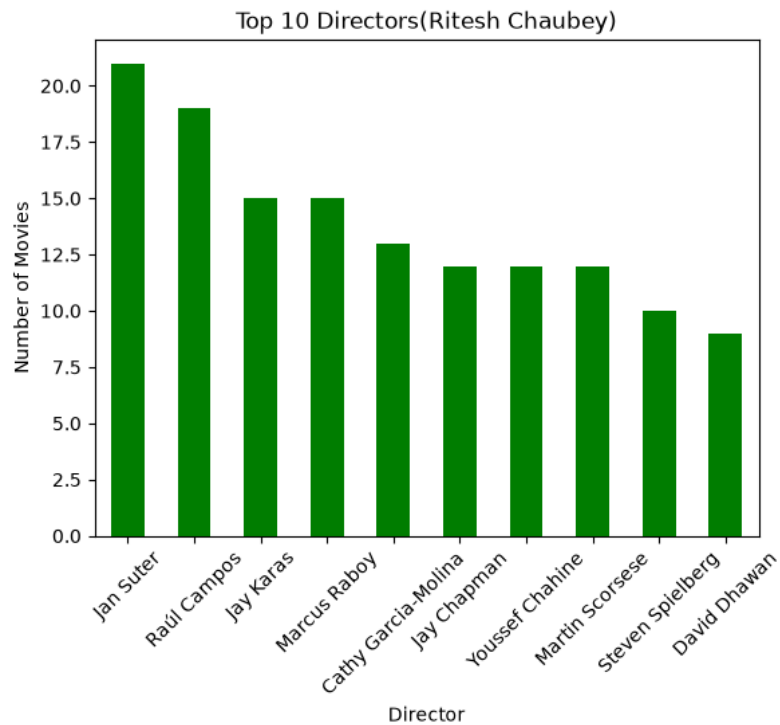
```
plt.show()
```

Output:-

```
Director with Highest Movies
Jan Suter - 21

Director with Highest TV Shows
Alastair Fothergill - 3

Top 10 Directors
director
Jan Suter                21
Raúl Campos              19
Jay Karas                15
Marcus Raboy             15
Cathy Garcia-Molina      13
Jay Chapman              12
Youssef Chahine          12
Martin Scorsese          12
Steven Spielberg         10
David Dhawan              9
Name: count, dtype: int64
```



Q.5 Compare Netflix releases:

- Before 2020.
- During 2020.
- After 2020.
- Display the result using a bar chart or line chart.

Code:-

```
# Count releases
```

```
before_2020 = len(df[df["release_year"] < 2020])
```

```
during_2020 = len(df[df["release_year"] == 2020])
```

```
after_2020 = len(df[df["release_year"] > 2020])
```

```
# Print the results
```

```
print("Netflix Releases Comparison")
```

```
print("-----")
```

```
print("Before 2020 :", before_2020)
```

```
print("During 2020 :", during_2020)
```

```
print("After 2020 :", after_2020)
```

```
# Bar Chart
```

```
labels = ["Before 2020", "During 2020", "After 2020"]
```

```
values = [before_2020, during_2020, after_2020]
```

```
plt.figure(figsize=(6,5))
```

```
plt.bar(labels, values, color=["blue", "orange", "green"])
```

```
plt.title("Netflix Releases Comparison")
```

```
plt.xlabel("Year Category")
```

```
plt.ylabel("Number of Releases")
```

```
plt.show()
```

Output:-

```
Netflix Releases Comparison
-----
Before 2020 : 6888
During 2020 : 868
After 2020  : 31
```

